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Development of a MALTA2-Based Fast MAPS Tracker: Detector Design, Readout Firmware, and Performance Validation

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Abstract Insert your abstract here. Insert your abstract here. Insert your abstract here. Insert your abstract here. Insert your abstract here. Insert your abstract here. Insert your abstract here. Insert your abstract here. Insert your abstract here.

1 Introduction - (lead by Xuan, Dien)

The Electron-Ion Collider (EIC) will provide high-luminosity, high-energy electron+proton ($e+p$) and electron+nucleus (e+A) collisions, enabling three-dimensional imaging of nucleons and nuclei, advancing our understanding of the proton spin puzzle, and investigating the origin of visible matter as well as the machanisms of hadron formation [1]. The EIC is designed to accommodate up to two Interaction Points (IPs): one for the project detector being developed by the ePIC collaboration and another reserved for a future second EIC detector. The collider will operate with a bunch crossing interval of 10.2 ns, while the Deep Inelastic Scattering (DIS) physics event rate is expected to be approximately 500 kHz rate. Consequently, the detector systems will be exposed to background particle rates that are substantially higher than those associated with DIS physics events, presenting significant challenges for detector design, readout, and event reconstruction. Recent experimental and theoretical studies for the EIC [2–7] have demonstrated that precise reconstruction of heavy-flavor hadrons and jets in the forward pseudorapidity region ($\eta > 2$) is essential for constraining nucleon and nuclear parton distribution functions (PDFs) at high Bjorken-x (x_{BJ}), as well as for elucidating heavy-quark hadronization mechanisms. These measurements are directly aligned with the primary scientific objectives of the EIC [1].

To support either a future upgrade of the ePIC detector or the second detector at the EIC, we are developing a forward silicon tracker based on the Depleted Monolithic Active Pixel Sensor (DMAPS) technology, specifically the MALTA2 sensor [8]: Fast MAPS Tracker (FMT). A comprehensive detector R&D program, together with detailed simulation studies, has been carried out to evaluate the detector performance and optimize its design. The primary objectives of this detector are: 1) to achieve fine spatial resolution and fast timing capability while maintaining reliable performance under the radiation doses from multiple years of EIC operation; 2) to improve charged-particle tracking performance in the forward pseudorapidity region; and 3) to provide sufficient time resolution to distinguish particles originating from consecutive bunch crossings separated by 10.2 ns, thereby mitigating beam-induced background contamination in physics events.

In this paper, we present the design and development of the fundamental building unit of the FMT, the quad-sensor module, together with updates to the readout firmware relative to the previous sensor readout system. We then present detailed simulation studies evaluating the detector's tracking performance, timing capability, and impact from EIC backgrounds in the forward region. These results provide valuable guidance for the design optimization of a future forward tracking detector at the EIC.

2 MALTA2 Technology - (Lead by Carlos)

The Tower 180 nm CMOS Imaging Sensor (CIS) technology, implemented in the MALTA family of Depleted Monolithic Active Pixel Sensors (DMAPS) [8, 9], offers several

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advantages that make it an attractive candidate for tracking detectors at the EIC. First, its monolithic architecture integrates the sensing element and front-end readout electronics on the same silicon substrate, eliminating the need for costly and material-intensive bump-bonding. This integration reduces both the detector cost and material budget. Second, this technology empolys small collection electrodes with an input capacitance below 5 fF, resulting in low noise (< 20 ENC), low power consumption (< 20 mW/cm^2), and fast signal formation with a response time below the EIC bunch-crossing interval of 10.2 ns.

The MALTA2 sensor used in this work is the second-generation device in the MALTA sensor family, with significant improvements in in-pixel charge collection efficiency and radiation tolerance. Each MALTA2 sensor comprises a matrix of 224×512 pixels, corresponding to an active area of 18.3 mm^2, and with a pixel pitch of 36.4 μm. The sensor integrates an analog front-end within each pixel that functions as a charge amplifier and discriminator prior to digitization. A hierarchical asynchronous data-merging architecture enables simultaneous transmission of hit information from across the entire matrix, providing fast readout and enhanced timing resolution.

The MALTA series sensors have demonstrated excellent performance in beam test [10], including fine spatial granularity, a time resolution better than 2 ns, a signal-to-noise ratio of approximately 20 for minimum-ionizing particle (MIP) signals, resulting in a low fake-hit rate, and high charge-collection efficiency. These characteristics are critical for precise vertex and tracking reconstruction in the high-rate EIC environment. Furthermore, the high-resistivity, fully depleted sensing volume enables efficient charge collection and good radiation tolerance, while the thin sensor design reduces multiple scattering, thereby improving both momentum and impact-parameter resolutions. Collectively, these features make the Tower 180 nm CMOS technology a compelling choice for constructing large-area, high-performance, and cost-effective silicon tracking systems for future EIC experiments.

3 Detector and Module Design - (lead by Lindsey, Kerem)

3.1 FMT geometry

To improve tracking performance in the forward pseudo-rapidity region ($\eta > 2.5$), the current Fast MAPS Tracker (FMT) design consists of two silicon tracking disks surrounding the beam pipe and positioned downstream of the central tracking system. The detector geometry has been optimized by balancing forward tracking acceptance with detector integration constraints. The longitudinal position of the first disk was optimized within the range of 140-150 cm from the interaction point, while the second disk was placed between 160 and 165 cm. The ePIC interaction region employs a 25 mrad beam-crossing angle, whereas the interaction region for the future second EIC detector has not yet been finalized. Consequently, the inner radius of the FMT is constrained by the current ePIC beam-pipe design. In the present design, both tracking disks have an inner radius of 7.012 cm and the outer radius of 23.012 cm. Fig 1 presents the geometry of the FMT implemented in the standalone simulation. The first and second detector disks are positioned at z = 145 cm and z = 165cm, respectively.

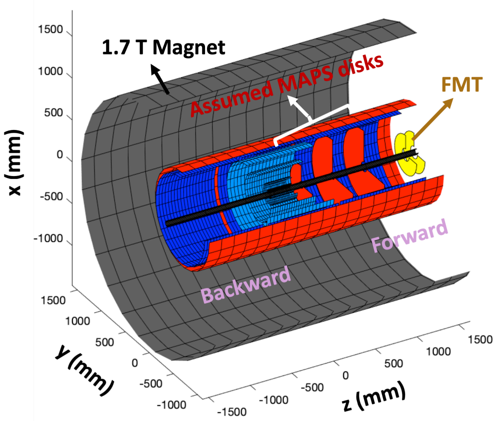

Fig. 1 Standalone simulation geometry showing the Fast MAPS Tracker (FMT, yellow) positioned downstream of the three MAPS tracking disks (red) within the ePIC solenoidal magnet (grey).

3.2 MALTA2 quad-sensor module prototype design

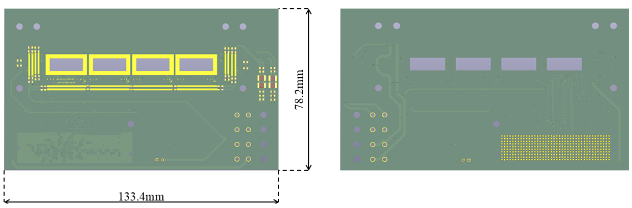

Fig. 2 3D rendering of FMT MATLA2 Quad Module FPC prototype. The front view is shown in the left and the back view is shown in the right.

The MALTA2 quad-sensor module is designed as the fundamental building block of the FMT. As shown in Fig 2, four MALTA2 sensors are mounted side by side on the prototype Flexible Printed Circuit (FPC). The readout and power pads are located along both sides of the FPC, enabling the interconnection of multiple sensors to construct a larger detector matrix while providing flexibility in the placement of the master chip. In the slave-master readout architecture, all detector data are transmitted to and read out through the master chip. The master chip collects data from the upstream slave chips and output the aggregated data stream through the Low-Voltage Differential Signaling (LVDS) pads located on its bottom side, as seen in Fig 3. This module design can minimize inter-chip communication distances while requiring only a single high-speed output link for the readout of the entire detector matrix. In the prototype carrier board used in this study, three slave chips are arranged in a row adjacent to the master chip, which is positioned on the left side of the module.

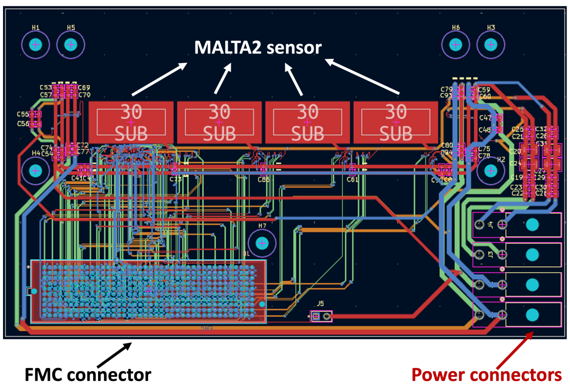

Fig. 3 Technical layout of the MALTA2 quad-sensor module prototype. The diagram illustrates the geometric arrangement of the four MALTA2 sensors, wire-bonding pads, and the routing paths for power distribution and high-speed differential signals.

The MALTA2 sensors are integrated onto a custom-designed FPC that provides the mechanical support and electrical interfaces required for tracker operation. Owing to the small bond-pad dimensions ($88\ \mu m \times 88\ \mu m$) and fine pad pitch on both the sensors and the FPC, reliable chip-to-chip and chip-to-board interconnections are critical for the assembly of the modular. The FPC layout was optimized to ensure precise alignment of the electrical and mechanical components. The MALTA2 sensors were attached to the FPC using aluminum wedge wire bonding performed at the CERN Bonding Laboratory. Figure 4 illustrates the wire-bonding scheme required for sensor operation. In this prototype, the leftmost sensor serves as the master chip, through which all detector readout and slow-control communication are routed, as reflected by the high density of wire bonds along its bottom edge.

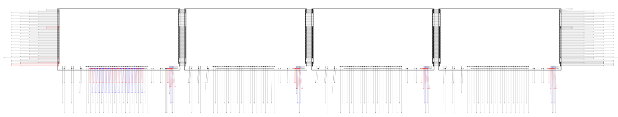

Fig. 4 Schematic of the wire-bonding layout for the MALTA2 quad-sensor module.

3.3 Initial Quad Module Testing

4 MALTA2 quad-sensor module readout architecture and firmware development - (lead by Joellen, Kerem)

4.1 Readout Architecture

The MALTA2 quad-sensor module utilizes 38 Low-Voltage Differential Signaling (LVDS) pairs originating from the bottom side of the master chip. Each MALTA2 sensor comprises of 256 double column of pixels, which form the basis of the hierarchical data-merging architecture within the pixel matrix. When a hit is registered, the corresponding data propagates toward the end of the column as a 22-bit word, where the data-merging process begins. A hierarchy of eight merging stages consolidates the hit information from all 256 double columns into a 37-bit output word for transmission through the LVDS links, including a stage that combines the positive- and negative-parity groups within each double column. CMOS transistors located along the left and right edges of each chip enable data to be forwarded from the three secondary chips to the master chip, where the data streams are combined for final readout. The propagation delays associated with transferring data from the secondary sensors to the master chip are accounted for through the appropriate timing calibration/synchronization procedure. After the successive inter-chip merging stages are completed, the resulting data are transmitted through the LVDS links to the FPGA for further processing.

4.2 Firmware Architecture

The readout of the MALTA2 quad-module is being developed using the Alinx AXKU040 (Kintex UltraScale) platform. The custom FPGA design is based on a previous version that used the KC705 (Kintex 7) platform for readout of a single MALTA2 chip. Firmware updates were introduced to support quad-module operation by extending the data-input and slow-control interfaces while retaining the existing asynchronous readout architecture.

For the quad-module implementation, the FPGA receives 38 differential signal pairs from the module. The additional information with respect to the single-chip readout provides the chip identification required to distinguish data originating from the four MALTA2 sensors. A two-bit chip identifier is associated with the data stream, with the values 00, 01, 10, and 11 identifying the four individual sensors. This allows data from all four MALTA2 chips to be transmitted through the common module readout path while preserving the identity of the sensor from which each hit originated.

The readout of the chips is fully asynchronous with no clock distribution needed between the chips and FPGA. Each half of the differential signals is asynchronously oversampled using separate Input Delay Elements (IDELAYE) and Input Serializer/Deserializers (ISERDES). Using 640 MHz DDR sampling together with the independently delayed P and N signal paths, the implemented oversampling scheme achieves an effective sampling rate of 2.56 Gb/s.

Moving from the Kintex 7 platform to the Kintex UltraScale platform required updating the asynchronous sampling module to use IDELAYE3/ISERDES3 elements instead of IDELAYE2/ISERDES2 elements.

Each MALTA2 sensor is configured through its dedicated slow-control interface, in which the configuration data are serially shifted into the on-chip configuration registers. For quad-module operation, separate slow-control data paths are provided for the four sensors, allowing the configuration of each chip to be programmed independently. This enables independent adjustment of the chip-level configuration parameters, including the programmable DAC settings, for each MALTA2 sensor.

5 Performance Validation in Simulation - (Lead by Abhishek, Amelia, Lucian, Vlad)

The charge collection of the MALTA2 pixel has been derived from testbeam data [11]. A parametric simulation approach is used that has been validated against data [12]. A quad module made of four sensors in series simulated in GEANT4. Each sensor is identical but has time difference because of proximity (ref simulated geometry and schemtic). Characterization of sensors: Each sensor physical properties. fast simulation and Readout implementation(digitization) and tracking with custom configuration.

5.1 Telescope geometry and setup in simulation

The key physical dimensions of the simulated geometry are summarized in Table 1.

The telescope is modelled in GEANT4 as a quad module of four identical MALTA2 sensors placed in series along the beam axis. Each sensor is a $1.8 \times 0.9\ cm^2$ monolithic CMOS pixel matrix with a pixel pitch of 36.4 µm, giving a 512×224 pixel grid. The sensors are separated by a fixed spacing, so a particle crosses the planes sequentially. Small time offsets between planes, arising from their relative position and readout propagation, are applied per sensor.

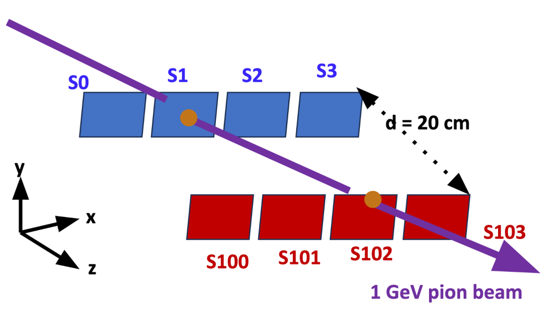

Fig. 5 Telescope geometry and setup in simulation.

Table 1 Key physical dimensions of the simulated telescope geometry.

Component	Parameter	Value
World volume	dimensions	$2\,\mathrm{m} \times 2\,\mathrm{m} \times 2\,\mathrm{m}$
	material	vacuum (`G4_Galactic`)
MALTA2 sensor	lateral size	$1.86368\,\mathrm{cm} \times 1.86368\,\mathrm{cm}$
	sensitive thickness	29.1 µm
	pixel pitch	36.4 µm
	pixel matrix	512×512
Charge-collection model	Gaussian σ	4.3 µm in both axes
Quad module	sensors per module	4
	sensor-to-sensor pitch	1.86369 cm
	tracking-plane separation	10 cm
GEANT4 production cuts	γ, e^-, e^+	1 µm

Primaries are generated per event with a configurable multiplicity. The first primary in each event is the signal (a 1 GeV π^+) incident perpendicular to the sensor plane; any additional primaries are background (10 MeV e^-). Each primary is placed at a random position within the active area and assigned a global time $t = N_{\mathrm{evt}}\, t_{\mathrm{veto}} + \delta t$, where N_{evt} is the event index and δt is a random intra-spill offset up to 2 µs. Signal and background therefore overlap in time and are distinguished only by their truth label, not by arrival time.

Energy deposition in each sensor is digitized with the parametric MALTA2 response (charge collection, threshold, and timing), and the resulting pixel hits are reconstructed per plane. Truth provenance (event ID and track ID) is propagated through digitization to allow exact signal-to-hit association for the efficiency and tracking studies that follow.

5.2 Time Resolution (no background)

Plot: Time resolution (compare: without background, with 1,2,3 particles per event) Validated with data.

5.3 Background rate

Explain the expected background at the location of the FMT (from Amelia's plots?):

– Select region of $1 \times 2\,\mathrm{cm}^2$ with highest hit rate.

- Show distribution of hits per 2 µs event
- Show PDG code of background particles. Is it primarily photons and electrons/ positrons?

Explain the simplifications/ assumptions that go into the MALTA2 simulation.

- Identify maximum expected background hit rate per sampled event.
- Spatial distribution is approximately flat within individual MALTA2 sensor.

5.4 Hit efficiency with background

Plot: Hit efficiency of a signal particle versus background rate per event:

- Position of background and signal is random in MALTA2 sensor.
- Keep time window of background particles within 10.2 ns.
- Increase number of background particles that fall within this. (Expectation from Lucian: Efficiency is still >90% for 1,2,3 background particles but decreases drastically for >3 particles.

5.5 Tracking efficiency, potential spatial/temporal resolution with background

This subsection can target the proper tracking efficiency using 2 FMT planes.

- Obtain hits per plane.
- Reconstruct a track from the two planes.
- Calculate pointing resolution of the track. In the simulation, signal is incident perpendicular onto the sensor. How is the direction reconstructed from the track?

5.6 Matching capability with background

6 Summary

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.

In some disciplines use of Discussion or 'Conclusion' is interchangeable. It is not mandatory to use both. Please refer to Journal-level guidance for any specific requirements.

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Appendix A: Appendix section

We consider a sequence of queueing systems indexed by n. It is assumed that each system is composed of J stations, indexed by 1 through J, and K customer classes, indexed by 1 through K. Each customer class has a fixed route through the network of stations. Customers in class k, $k = 1,\ldots,K$, arrive to the system according to a renewal process, independently of the arrivals of the other customer classes. These customers move through the network, never visiting a station more than once, until they eventually exit the system.

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